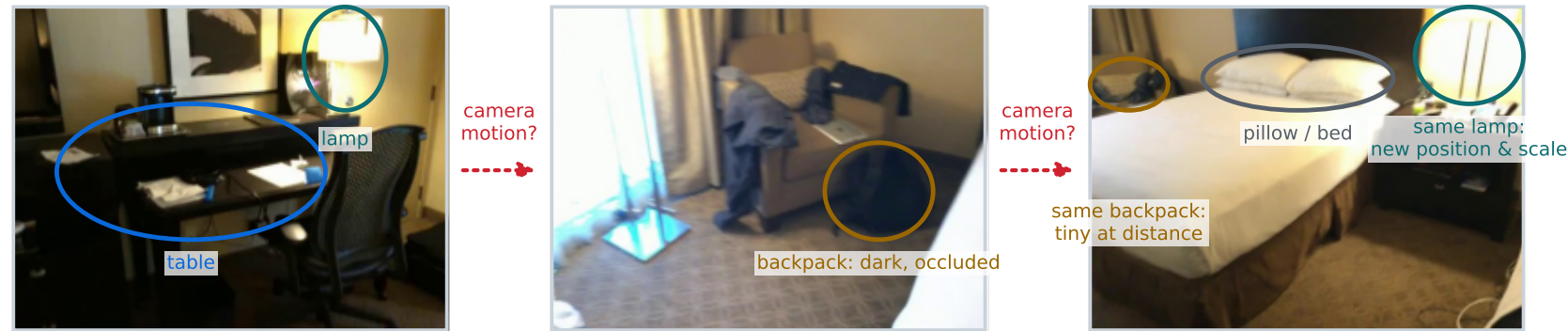


Q: "Which of these objects (backpack, pillow, bed, lamp) is the closest to the table?"

(a) Frame prompting: the evidence is scattered across viewpoints



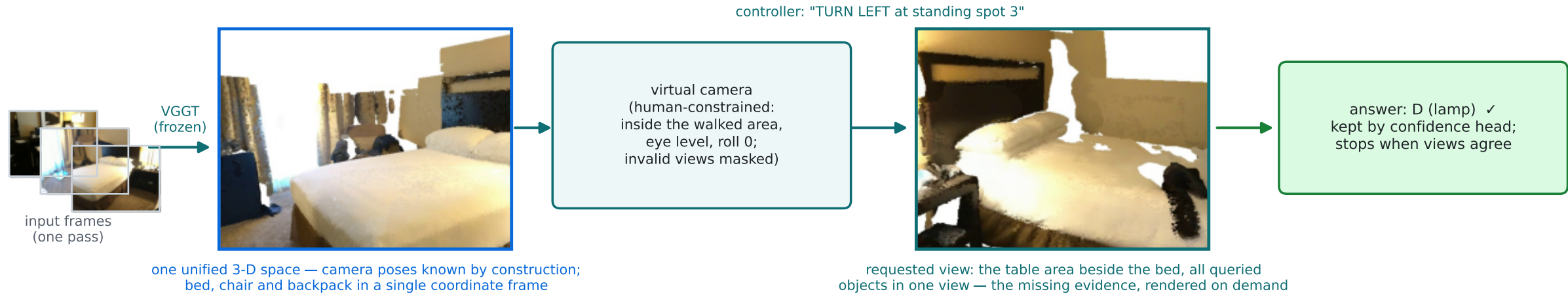
To answer, the VLM must implicitly:

- estimate camera motion between frames
- re-identify objects across views
- place everything in one 3-D layout
- ignore redundant / distracting views

—unreliable for (small) VLMs ✗

frames arrive as unrelated 2-D projections: each is encoded in its own camera coordinates; the table and the backpack never appear together

(b) ViewTree: reconstruct once into an explicit 3-D memory, then choose what to observe



reconstruction runs once per scene; the model explores by moving a virtual camera under physical constraints, acquires only the views it needs (mean < 5 VLM calls), and falls back to the direct answer when the memory adds nothing